

Sarang Nagpal

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Summary

Results-oriented Engineer and current Master's student with comprehensive experience building high-throughput data systems, scalable architectures, and resilient AI infrastructure. Proven ability to leverage generative AI models, deep learning frameworks, and modern tools to solve real-world business challenges, including managing AI inference engines that handle more than a Billion transactions every day. Highly adept at system design, utilizing modern DevOps practices, orchestrating cloud-native deployments, and strategically applying data analytics to solve both internal and external business problems at enterprise scale.

Work Experience

Visa Inc.

Data Engineer

April 2025 - August 2026

Bangalore, India

- Engineered a token bucket-based rate limiter within a high-performance Go-based distributed system and load generator deployed in Kubernetes, enabling precise control of stress test throughput at more than 100,000 Transactions per second to improve system scalability.
- Initiated and established foundational practices for chaos engineering in AI inference services used for Risk Modeling of financial transactions, focusing on proactively injecting both kubernetes and vanilla cloud infrastructure faults to validate system resilience.
- Designed test cases for multiple components of the architecture simulating latency, resource exhaustion, and stale context scenarios specifically to test the robustness of the Risk Modeling AI inference engine.
- Developed a tool agnostic Jenkins based cron system to automate the execution of chaos tests and generate detailed reports on system performance, enabling continuous improvement of AI inference reliability.
- Collaborated cross-functionally with data scientists and architects to develop telemetry metrics and observability dashboards in Prometheus and Grafana, accurately measuring the blast radius of chaos tests on AI inferencing.
- Optimized and fine-tuned kubernetes distributed Golang and Kafka data pipelines to improve throughput ensuring the system meets stringent SLAs for real-time financial transaction processing.
- Improved platform availability towards a 99.9999%ile target by contributing to distribution of infrastructure into independent operational stacks spread across multiple Availability Zones, ensuring high availability and disaster recovery for critical financial services.

Amazon

Software Development Engineer

January 2024 - February 2025

Bangalore, India

- Improved the Unified Payments Interface (UPI) Payment Method onboarding Success Rate by 5% for an active base of over 20 million customers by shortening the onboarding process and reducing the number of clicks required for new registrations.
- Engineered a system to securely register existing UPI customers across multiple payment service providers without scaling onboarding services, mitigating single point of failure issues.
- Designed and developed robust RESTful APIs utilizing Java alongside a suite of serverless AWS components, including AWS Lambda, DynamoDB, SQS, and SNS.
- Modernized legacy mobile web interfaces by upgrading from JavaServer Pages (JSP) to ReactJS with Redux, resulting in an 8% improvement in digital issuance metrics and decreased latency.

Amazon

Software Development Engineer Intern

May 2022 - July 2022

Bangalore, India

- Conceptualized and built a standalone backend service utilizing Spring Boot (Java) to load test a core UPI service.
- Scaled EC2 instance infrastructure dynamically to successfully execute custom stress testing frameworks in a simulated high-concurrency environment.
- Achieved a 100% score in the resilience metric by measuring and documenting performance thresholds prior to the conclusion of the internship.

Education

Master of Science

Computer Engineering

RWTH Aachen University

May 2026 - Present

Bachelor of Technology

Electronics and Communication Engineering

National Institute of Technology, Warangal

May 2019 - June 2023

Developed a strong mathematical foundation in control systems, advanced physics, low-level algorithmic design, and semiconductor device physics and design.

Skills

Programming

Go (Golang), Python, C++, Java, JavaScript, TypeScript, SQL

AI & Data Analytics

GenAI, Large Language Models (LLMs), TensorFlow, Keras, Pandas, Prompt Engineering, Data Aggregation

Cloud & DevOps

AWS (Lambda, DynamoDB, SQS, SNS), Kubernetes, Docker, ChaosMesh, Kafka, RabbitMQ, Jenkins

System Design

Microservices, Serverless Architecture, Event-Driven Messaging, RESTful APIs, Distributed Systems, High-Throughput Data Pipelines

Languages

English (C2 - IELTS 8.0), German (A2 - Goethe Zertifikat)

Certifications

Certified Kubernetes Application Developer (CKAD)

Cloud-Native Architecture Certification

The Linux Foundation

August 2026

Hands-on competence in designing, building, configuring, and deploying complex cloud-native applications directly for distributed Kubernetes environments.

Machine Learning Certification

Algorithmic Foundation Certification

Stanford University (Coursera)

December 2022

Mathematical foundation of supervised and unsupervised learning.

Deep Learning Specialization

Applied Neural Networks Certification

Andrew Ng (Coursera)

September 2020

Convolutional Neural Networks (CNNs); Sequence Models; hyperparameter tuning; mathematical regularization; optimizing deep networks.